Comparative Analysis of PCA and MDS for Dimensionality Reduction in Labeled Datasets

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Abstract

High-dimensional data analysis relies heavily on Dimension Reduction (DR) techniques to project complex datasets into visualizable 2D or 3D spaces. However, these projections inevitably introduce topological distortions. This study presents a Visual Analytics dashboard designed to evaluate and compare the fidelity of Principal Component Analysis (PCA) and Euclidean Multidimensional Scaling (MDS). By integrating quantitative metrics such as Trustworthiness, Continuity, and local proximity indices (Precision and Recall), the system identifies projection-induced false positives and false negatives. Furthermore, the introduction of a 2D-perspective view utilizing K-Means clustering and Sankey diagrams enables the tracking of the "Cluster Agreement Flow", highlighting how algorithmic choices directly influence the interpretation of natural data groupings.

1 Introduction

Dimensionality reduction is an essential step for the visual exploration of complex datasets; however, every projection operates on an inherent trade-off between global distance preservation and local neighborhood fidelity. Consequently, the evaluation of DR techniques cannot rely solely on visual inspection but requires rigorous quantitative analysis. To investigate these projection behaviors across varying topological structures, our analysis incorporates three distinct datasets: Wine Quality, Iris, and User Knowledge.

Our primary objective is the creation of a comprehensive Visual Analytics tool intended for domain experts. This system is engineered to rigorously evaluate dimensionality reduction projections and to serve as a foundational instrument for designing and validating new Visual Analytics platforms. To achieve this, we aim to transcend purely aesthetic representations by explicitly exposing structural distortions:

- **False Positives (Precision):** Points that are distant in the original high-dimensional space but appear proximally close in the 2D projection, thereby creating artificial and misleading clusters.
- **False Negatives (Recall):** Inherently similar points belonging to the same class that are projected far apart, fragmenting the intrinsic structure of the data.

The proposed system juxtaposes the linear, variance-maximization approach of Principal Component Analysis (PCA) against the pairwise distance-preserving approach of Euclidean Multidimensional Scaling (MDS). By doing so, it quantifies the extent to which a 2D projection accurately represents the original multidimensional space, thereby providing a dynamic visual judgment of the underlying dataset.

2 Technologies Used

The technical architecture employs a decoupled model, utilizing Python for intensive data preprocessing and web standards for real-time interactive visualization.

- **Backend and Data Processing:** The analytical pipeline is heavily supported by the Python ecosystem, specifically relying on the `scikit-learn` and `scipy` libraries for machine learning, clustering, and dimensionality reduction tasks. The key algorithms utilized include:
  - `sklearn.preprocessing.StandardScaler` to standardize the raw N-dimensional features, ensuring uniform scale and variance.
  - `sklearn.decomposition.PCA` and `sklearn.manifold.MDS` for computing the 2D spatial projections.
  - `sklearn.cluster.KMeans` for extracting data-dependent clusters in both N-dimensional and 2D-projected spaces.
  - `scipy.optimize.linear_sum_assignment` (implementing the Hungarian Algorithm) to compute the optimal bipartite matching between the predicted K-Means clusters and the abstract ground-truth labels.
- **Frontend and Interactive Visualization:** The user interface is a Single Page Application (SPA) driven by the `D3.js` (Data-Driven Documents) library. The frontend utilizes D3.js to manipulate the Document Object Model (DOM) by binding data directly to Scalable Vector Graphics (SVG). This enables the creation of custom

visualizations such as the PCA and MDS scatterplots, the interactive Parallel Coordinates view, and the Sankey diagram for tracking cluster flows. To provide a responsive layout and UI components—such as the dataset selection models and dynamic filtering control panels—a custom CSS Grid architecture was developed, ensuring accessibility across different device screen sizes and supporting a dynamic light/dark theme transition.

- **Data Format and Communication:** Data exchange between the preprocessing backend and the frontend is standardized using JSON (JavaScript Object Notation). This format is ideal for web-native visualizations because it is natively parsable by D3.js and represents complex multi-dimensional data efficiently. To maintain a stable visual scale across different filtered views, the system caches a global version of the dataset in memory, ensuring that radar chart axes and scatterplot coordinate domains remain strictly consistent during dynamic user exploration and interactive brushing.

3 Dataset and Preprocessing

To comprehensively evaluate the behavior of Dimensionality Reduction (DR) techniques across diverse topological structures, the system utilizes three well-known multivariate datasets. The preprocessing and metric computation pipeline is meticulously designed to evaluate both local neighborhood preservation and global structural fidelity.

3.1 Datasets Description

The visual analytics platform processes the following structured datasets:

- **Wine Quality Dataset:** Serving as the primary case study, this dataset consists of 178 instances of wine originating from three different cultivars (abstract classes). It features 13 continuous numeric attributes representing chemical constituents (e.g., alcohol, malic acid, flavonoids).
- **Iris Flowers Dataset:** A foundational dataset containing 150 instances evenly distributed across three species of Iris flowers (50 instances each). It comprises 4 numeric attributes (sepal and petal dimensions).
- **User Knowledge Modeling Dataset:** This dataset contains 403 instances representing students' knowledge status about the subject of Electrical DC Machines. It includes 5 numeric input attributes and is categorized into 4 distinct knowledge levels (e.g., Very Low, Low, Middle, High).

3.2 Standardization

Euclidean distance is highly sensitive to the magnitude of the underlying variables.

To prevent features with larger scales from disproportionately dominating the distance calculations, the first step in the pipeline (`compute_relationship.py`) standardizes all numeric variables. We employ Z-score normalization via `scikit-learn`'s `StandardScaler`:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where μ is the mean of the feature and σ is its standard deviation.

3.3 Local Quality Metrics: Precision, Recall, and F-Score

Recent literature emphasizes that relying solely on visual inspection or global metrics is insufficient for understanding why expected cluster structures disappear in projections. As formalized by van der Hoorn et al. in *"Why Can't I See My Clusters? A Precision-Recall Approach to Dimensionality Reduction Validation"* [7], DR validation must explicitly assess the relationship phase by quantifying localized false positives and false negatives.

In our system, these metrics are implemented within the `compute_relationship.py` script. First, a K-Nearest Neighbors (K-NN) graph is constructed in the high-dimensional space ($K = 15$). For a given data point i belonging to a true abstract class $C(i)$, let $N_k(i)$ represent the set of its k nearest neighbors, and let D represent the entire dataset. The script computes:

- **Local Precision (Lack of False Positives):** Measures the proportion of projected neighbors that actually belong to the same true class. It penalizes points that are distant in the original space but appear proximally close in the projection.

$$\text{Precision}(i) = \frac{|\{j \in N_k(i) \mid C(j) = C(i)\}|}{|N_k(i)|} \tag{2}$$

- **Local Recall (Lack of False Negatives):** Measures the proportion of points from the same original class that are successfully retained within the local neighborhood, penalizing instances where inherently similar points are projected far apart.

$$\text{Recall}(i) = \frac{|\{j \in N_k(i) \mid C(j) = C(i)\}|}{|\{j \in D \setminus \{i\} \mid C(j) = C(i)\}|} \tag{3}$$

- **Local F-Score:** Provides a unified assessment of topological fidelity by computing the harmonic mean of Local Precision and Local Recall. This metric balances the trade-off between false positives and false negatives for each individual point.

$$\text{F-Score}(i) = 2 \cdot \frac{\text{Precision}(i) \cdot \text{Recall}(i)}{\text{Precision}(i) + \text{Recall}(i)} \tag{4}$$

These localized continuous metrics are exported in JSON format to dynamically drive the frontend's color mapping, enabling users to visually isolate structural anomalies directly on the 2D scatterplots.

3.4 Global Quality Metrics: Trustworthiness and Continuity

To complement the local point-wise evaluation, the system computes global distance preservation metrics, adhering strictly to the methodologies highlighted in the academic survey *"Toward a Quantitative Survey of Dimension Reduction Techniques"* [6]. These metrics evaluate the overall rank preservation of distances before and after the projection.

Within the `compute_projection.py` script, the pipeline utilizes the `sklearn.manifold.trustworthiness` module to calculate:

- **Trustworthiness (T):** Penalizes false positives. If a point j is in the K-neighborhood of i in the 2D projected space but not in the original N-dimensional space, the projection's trustworthiness decreases. It is mathematically defined as:

$$T(K) = 1 - \frac{2}{NK(2N-3K-1)} \sum_{i=1}^{N} \sum_{j \in U_i(K)} (r(i,j) - K) \quad (5)$$

  where $r(i,j)$ is the rank of point j based on pairwise distances in the original space, and $U_i(K)$ is the set of points that are in the K-neighborhood of i in the 2D space but not in the original space.

- **Continuity (C):** Conversely, Continuity penalizes false negatives, occurring when points that are neighbors in the original multidimensional space are pushed apart in the 2D projection. It is mathematically defined as:

$$C(K) = 1 - \frac{2}{NK(2N-3K-1)} \sum_{i=1}^{N} \sum_{j \in V_i(K)} (\hat{r}(i,j) - K) \quad (6)$$

  where $\hat{r}(i,j)$ represents the rank of point j based on pairwise distances in the 2D projected space, and $V_i(K)$ constitutes the set of points that are in the K-neighborhood of i in the original space but are missing from the K-neighborhood in the 2D projection.

These global metrics are computed for both the PCA and MDS projections and are prominently displayed in the dashboard's footer, providing domain experts with an immediate, baseline quantitative comparison of the projections' structural preservation capabilities.

4 Comparative Methodology: PCA vs MDS

To rigorously evaluate the projection artifacts exposed by the visual analytics dashboard, it is imperative to understand the theoretical foundations, objective functions, and expected behaviors of the dimensionality reduction algorithms under investigation. The system explicitly juxtaposes Principal Component Analysis (PCA) and Multidimensional Scaling (MDS), as they represent two fundamentally different paradigms of structural preservation: variance maximization versus distance preservation.

4.1 Principal Component Analysis (PCA)

PCA is a deterministic, linear transformation technique that seeks to project high-dimensional data onto a lower-dimensional subspace while retaining as much of the original data's variance as possible.

Mathematically, given a column-standardized dataset matrix $X \in \mathbb{R}^{N \times M}$ (where N is the number of samples and M is the number of features), PCA computes the covariance matrix Σ:

$$\Sigma = \frac{1}{N-1} X^T X \quad (7)$$

The algorithm then solves the eigenvalue problem for Σ:

$$\Sigma v = \lambda v \quad (8)$$

where λ represents the eigenvalues (amount of variance explained) and v represents the eigenvectors (the principal components). The 2D projection is obtained by multiplying the original data by the matrix W containing the eigenvectors corresponding to the two largest eigenvalues:

$$Y_{PCA} = XW \quad (9)$$

Advantages: PCA is computationally highly efficient ($O(M^3)$ for the eigendecomposition) and guarantees a global optimal solution without the risk of converging to local minima. It excels at capturing the global structure and dominant linear trends of the dataset.

Disadvantages and Expected Behavior: Because PCA is strictly linear and prioritizes global variance, it often fails to capture non-linear manifolds. In the context of visual analytics and clustering, PCA is highly susceptible to the *crowding problem*. By focusing on orthogonal variance, it frequently projects points from different classes into the same visual proximity (generating **False Positives**), thereby severely degrading the local *Precision* metric and *Trustworthiness*.

4.2 Multidimensional Scaling (MDS)

In contrast to PCA, Multidimensional Scaling represents a family of topological algorithms designed to map high-dimensional data into a lower-dimensional space such that the pairwise distances (or dissimilarities) between data points are preserved as faithfully as possible.

Generally, given an $N \times N$ dissimilarity matrix D where d_{ij} is the distance between points i and j in the original space, MDS seeks to find lower-dimensional coordinates $Y \in \mathbb{R}^{N \times 2}$ that minimize a cost function known as *Stress* (σ). The raw stress function is typically defined as:

$$\sigma(Y) = \sum_{i<j} (d_{ij} - \|y_i - y_j\|_2)^2 \quad (10)$$

Advantages: MDS is highly flexible; depending on the distance metric provided (e.g., Euclidean, Manhattan, or Geodesic for Isomap), it can capture non-linear relationships. It generally provides a much more faithful representation of local neighborhoods and class cohesion.

Disadvantages and Expected Behavior: Metric MDS optimization is computationally expensive (typically $O(N^3)$ per iteration using the SMACOF algorithm) and scales poorly to massive datasets. Furthermore, iterative optimization can become trapped in local minima. However, regarding spatial fidelity, it is expected to preserve intra-class relationships far better than PCA, yielding higher *Recall* and *Continuity* scores.

5 Visualizations

The Visual Analytics system is built upon a dynamic, highly coordinated dual-tab architecture designed to address the core challenge of dimensionality reduction: distinguishing between the intrinsic topological structure of the data and the artificial distortions introduced by projection algorithms. The interface is partitioned into two main analytical perspectives: the *nD Analysis* tab, which evaluates projection quality against the original high-dimensional ground truth, and the *2D Analysis* tab, which focuses entirely on comparing the geometric and categorical outcomes of the two embedded spaces.

5.1 Coordinated Projection Spaces

The core of the dashboard consists of three synchronized scatterplots that provide an immediate visual juxtaposition of the algorithms. The first two plots render the Principal Component Analysis (PCA) and Euclidean Multidimensional Scaling (MDS) embeddings side-by-side. This arrangement allows analysts to visually contrast the linear, variance-maximizing approach of PCA against the non-linear, pairwise distance-preserving approach of MDS.

To provide a structural baseline, the third scatterplot visualizes data-dependent groupings generated via K-Means clustering. In the nD view, this plot renders the "true" multidimensional clusters projected onto a 2D canvas, whereas in the 2D view, it displays clusters computed directly on the projected coordinates. This design choice explicitly highlights how algorithmic reduction alters the interpretation of natural data groupings.

5.2 Multivariate Contextualization

Because dimensionality reduction inherently forces a loss of information, evaluating 2D points in isolation is insufficient. To recover this lost context, a Parallel Coordinates Plot (PCP) is anchored to the lower section of the interface, displaying the complete multivariate profile (e.g., Alcohol, Flavanoids, Color Intensity) of the dataset. This component serves a crucial analytical purpose: it allows users to trace spatial outliers or dense 2D clusters back to their original high-dimensional signatures, revealing which specific features drive the formation of visual groupings or the occurrence of projection artifacts.

5.3 Structural Discrepancy Flow (Sankey Diagram)

While scatterplots are effective for spatial overview, they fail to explicitly quantify how individual instances migrate between categories across different algorithms. To solve this, the 2D Analysis tab introduces a bipartite Sankey diagram labeled *Cluster Agreement Flow*. This visualization maps the trajectory of data points from the K-Means clusters computed in the PCA space to those computed in the MDS space.

To overcome the "visual scarcity" problem—where a high percentage of agreement visually overwhelms the few discrepancies—the visualization encodes topological mismatch heavily. Links representing agreements are rendered in a subtle, semi-transparent gray, while discrepancies (points that shift cluster assignment between PCA and MDS) are rendered in solid, thick red bands. This ensures that even a single misclassified point remains highly salient and immediately draws the analyst's attention.

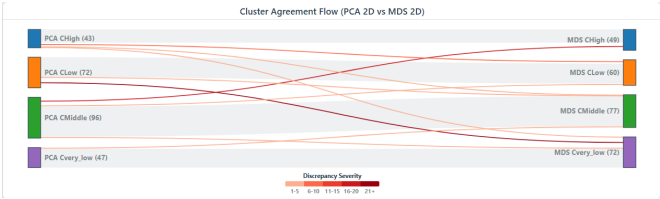


Figure 2: The Cluster Agreement Flow (Sankey diagram) quantifying the topological divergence between PCA and MDS. Red pathways explicitly highlight data points that received conflicting cluster assignments across the two projection spaces.

5.4 Local Diagnostic Analytics

To facilitate the transition from global overview to granular inspection, the rightmost section of the dashboard hosts a *Live Analytics* panel that dynamically adapts to user input:

- **Quality Gauges:** Real-time radial gauges display the average Precision, Recall, and F-Score of the current selection, quantifying the preservation of the k-Nearest Neighbor (k-NN) relationships.

- **Neighbor Graph:** A force-directed node-link diagram that reconstructs the true high-dimensional neighborhood of a selected point. This visualizes projection-induced *false positives* (distant points projected too closely) and *false negatives* (close neighbors scattered too far apart).

- **Radar Chart:** When a single point is selected in the 2D tab, a multidimensional radar chart compares the point's specific feature vector against the average profile of its ground-truth class and its assigned K-Means cluster, providing chemical/physical justification for its spatial placement.

Figure 1: The main interface of the Visual Analytics dashboard. The layout integrates coordinated scatterplots for spatial comparison, Parallel Coordinates for multivariate context, and a dynamic panel for local diagnostics.

- **Confusion Matrices:** Automatically generated grids that cross-tabulate ground truth labels against data-dependent cluster assignments, offering a rigorous numerical breakdown of classification errors.

6 Interactions

The system's interaction model is heavily grounded in Shneiderman's mantra—*Overview first, zoom and filter, then details-on-demand*. The interactive elements within the dashboard are not merely navigational aids; they act as precise analytical instruments designed to dissect the projection space, allowing analysts to transition seamlessly from macro-level topological overviews to micro-level point diagnostics.

6.1 Coordinated Brushing, Linking, and Single-Point Isolation

All spatial projections, multivariate views, and analytical panels are strictly coupled through bidirectional linking. The dashboard supports two primary modes of spatial selection:

- **Rectangular Brushing:** Executing a brush operation on any of the scatterplots instantly triggers corresponding highlights across the entire dashboard. Selected points are maintained at full opacity, while non-selected points are heavily dimmed to reduce visual clutter. This enables analysts to isolate a dense visual cluster in the PCA space and instantly observe whether that same cohort remains cohesive in the MDS space, or if its multidimensional profile in the Parallel Coordinates Plot (PCP) reveals it to be a false aggregation.
- **Single-Point Selection:** Clicking an individual point triggers an exhaustive local diagnostic state. In the nD view, this action not only isolates the point but explicitly draws its true k-NN connections across the scatterplots, instantly revealing projection-induced false positives (intrusions). Simultaneously, the Live Analytics panel updates to display the force-directed Neighbor Graph. In the 2D view, single-point selection triggers the Radar Chart, enabling an immediate chemical or physical comparison against cluster centroids.

6.2 Metric-Driven Filtration and Semantic Encoding

Visual inspection of scatterplots is inherently susceptible to human bias, such as perceiving false clusters due to point overplotting. To counter this, the dashboard integrates interactions driven entirely by local quantitative metrics. Using the semantic color switcher, analysts can smoothly transition the visual encoding of the data points from their original categorical ground-truth labels to continuous color gradients representing local projection quality (Precision, Recall, or F-Score).

Furthermore, dual range sliders allow for real-time dataset filtration based on minimum acceptable thresholds for these metrics. Unlike simple opacity adjustments, adjusting these sliders physically removes sub-threshold points from the rendering pipeline (across scatterplots, the PCP, and the Sankey diagram). This allows the user to dynamically strip away

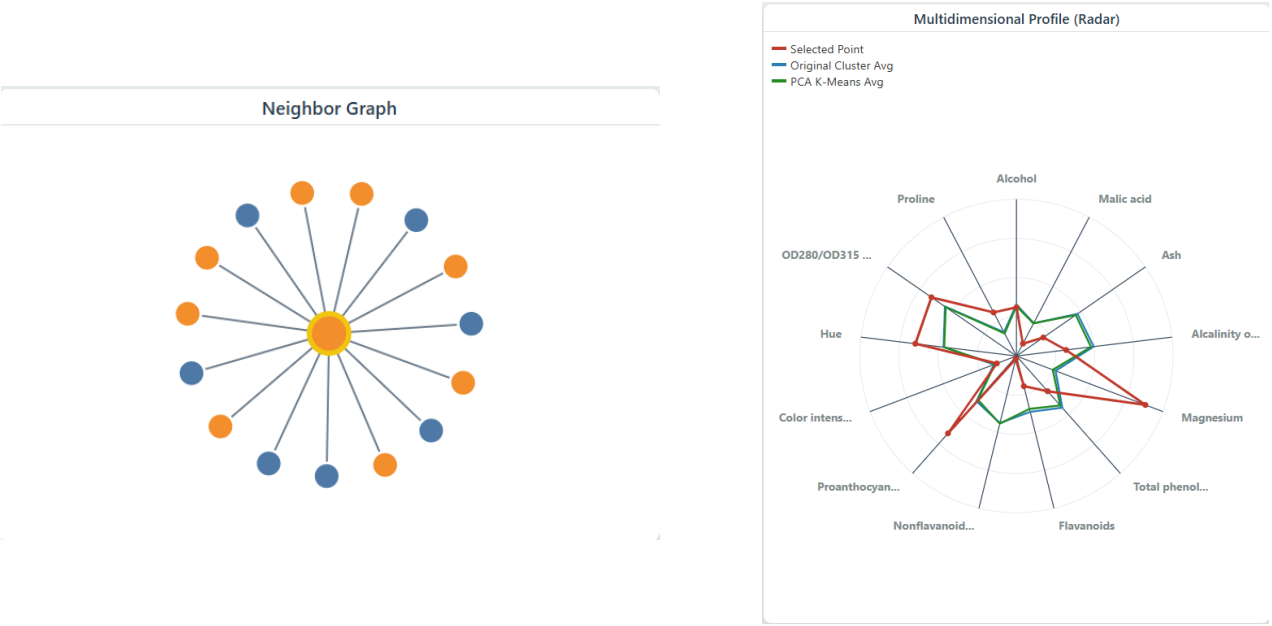


Figure 3: Dynamic *Details-on-Demand* visualizations. Left: A force-directed Neighbor Graph revealing local neighborhood fidelity. Right: A Radar Chart comparing a single selected instance against cluster centroids.

well-projected, reliable data points, effectively isolating and exposing only the most severely distorted regions of the manifold.

6.3 Cross-Space Navigation via Flow Selection

The Sankey diagram (*Cluster Agreement Flow*) functions as an active cross-filtering tool rather than a static chart. Because the diagram maps the movement of points between algorithmic categorizations, clicking on a specific flow pathway—such as a thick red link representing points misclassified between PCA Cluster 0 and MDS Cluster 1—acts as a complex conditional filter. This interaction instantly isolates those exact data points within the scatterplots and the PCP. It effectively bridges the gap between abstract categorical disagreement and spatial geometry, allowing the user to visually locate exactly where algorithmic confusion is physically occurring on the 2D plane.

6.4 Visualizing Centroid Anomalies and Matrix Synchronization

To explicitly expose the "gravitational pull" of incorrect clusters, users can activate the *Show Centroid Anomalies* toggle. This interaction calculates the geometric centroids of the K-Means clusters and overlays trajectory lines connecting data points to their algorithmically assigned centers. Correctly classified points are connected via faint, dashed gray lines, while structural anomalies (misclassified points) are connected via bold, solid red vectors. This feature transforms abstract classification errors into explicit visual geometries, immediately highlighting boundary regions where the 2D topology fails to cleanly separate data classes.

Additionally, users can dynamically toggle the *K-Means Source* (switching the nD K-Means plot to project onto either PCA or MDS coordinates) to test how different spatial geometries distort the structural cluster shapes. Activating the anomalies toggle also automatically synchronizes the Live Analytics panel to display the Confusion Matrices, providing a rigorous numerical breakdown alongside the visual trajectory lines.

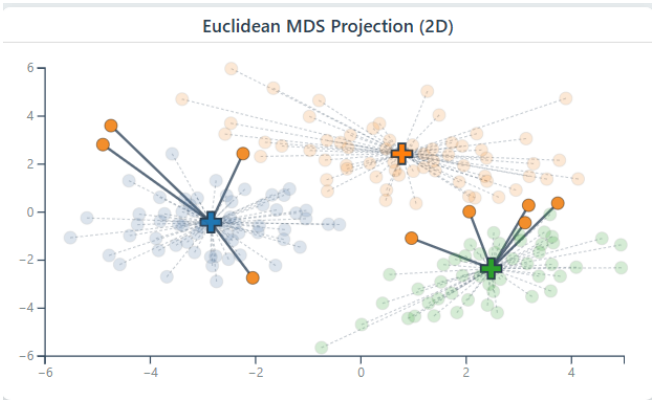


Figure 4: Visualizing centroid anomalies. Bold trajectory lines explicitly expose misclassified points, revealing the boundary regions where projection algorithms distort class separability. Simultaneously, the confusion matrix quantifies these visual discrepancies.

6.5 On-Demand Tooltips and Raw Feature Inspection

For the final level of detail, hovering over any point in the scatterplots summons an exhaustive, context-aware tooltip. It provides an immediate readout of the instance's algorithmic

health, including its exact Precision, Recall, and F-Score percentages, as well as distinct warnings if the point is suffering from severe False Positive (class mixing) or False Negative (cluster disconnection) distortions. To bridge the gap back to the raw data, hovering over the lines in the Parallel Coordinates Plot or the nodes in the Neighbor Graph displays the exact original multidimensional attributes (e.g., the specific concentrations of alcohol or flavanoids), ensuring the analysis remains grounded in the dataset's native domain.

7 Results and Insights

The deployment of the Visual Analytics dashboard across three distinct datasets—Iris, Wine Quality, and User Knowledge—yielded critical insights into the behavior of dimensionality reduction algorithms. By integrating local k-NN metrics (Precision, Recall, F-Score), the Parallel Coordinates Plot (PCP), and the *Cluster Agreement Flow*, we established a direct correlation between high-dimensional feature distributions and 2D algorithmic stability.

7.1 Global Metrics Evaluation Across Datasets

To empirically validate the theoretical differences between variance maximization (PCA) and distance preservation (Euclidean MDS), we conducted a comparative analysis of the global quality metrics across the three selected datasets. As formally defined in the methodology, Trustworthiness penalizes projection-induced false positives, while Continuity penalizes false negatives. Furthermore, the Global Relationship Matching score represents the dataset's average Local F-Score, providing a baseline indicator of intrinsic class cohesion within the local high-dimensional neighborhoods prior to specific algorithmic distortions.

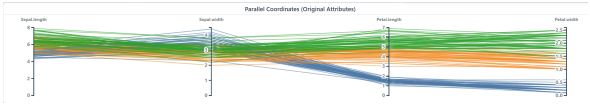
Metric	Wine Quality	Iris Flowers	User Knowledge
PCA Trustworthiness	0.897	0.983	0.820
PCA Continuity	0.943	0.992	0.921
MDS Trustworthiness	0.919	0.985	0.857
MDS Continuity	0.943	0.992	0.916
Relationship Matching (F-Score)	0.385	0.422	0.224

Table 1: Comparison of Global Quality Metrics Across Datasets

7.2 Multivariate Drivers of Projection Quality

The inclusion of the Parallel Coordinates Plot (PCP) provided the necessary high-dimensional context to explain *why* these 2D topological distortions occur. The visual patterns in the PCP directly predetermine the success or failure of the spatial embeddings:

- **Distinct Feature Bands ensure 2D Consensus:** When the PCP displays clear, non-overlapping routing of lines across specific axes (e.g., *Petal.Length* and *Petal.Width* in the Iris dataset), the resulting 2D projections are exceptionally stable. Both PCA and MDS easily capture these dominant variances, leading to near-perfect F-Scores and total algorithmic consensus.



- **Hybrid Profiles cause Boundary Distortions:** For misclassified or low-scoring boundary points (as seen in the Wine dataset), the PCP reveals "hybrid" feature profiles. A point that belongs to Class A but is projected into Class B's territory often shows line routings in the PCP where certain features (e.g., *Ash* or *Malic Acid*) physically intersect and mimic the profile of Class B. The projection algorithms mathematically average these conflicting features, resulting in spatial misplacements.

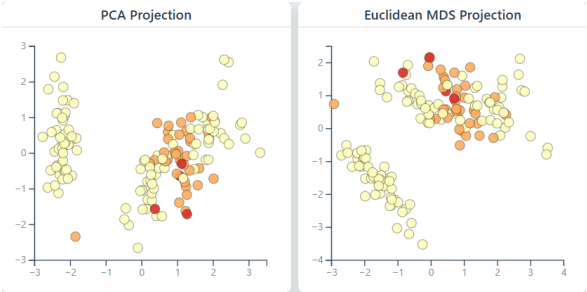
- **High-Dimensional Entanglement ensures 2D Chaos:** When the PCP resembles a "visual hairball"—a dense, chaotic intersection of lines across all dimensions with no separable bands (as seen in the User Knowledge dataset)—the 2D projections inevitably fail. The lack of separable variance in the original space forces the algorithms into arbitrary geometric compromises, destroying local neighborhoods and rendering any observed 2D clusters as mathematically unreliable artifacts.

7.3 Algorithmic Consensus on Separable Manifolds

Synthesizing these insights, datasets with clear intrinsic boundaries (Iris) yield nearly identical topological structures in both PCA and MDS. The scatterplots visually confirm the isolation of classes, the PCP shows distinct feature routing, and the Sankey diagram displays near-perfect parallel flows. This consensus validates that when a manifold possesses strong structural integrity, linear and non-linear reduction algorithms yield reliable and identical cluster definitions.

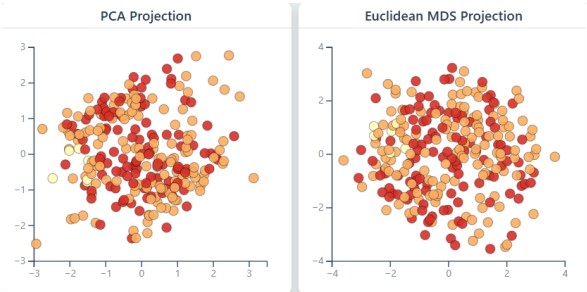
7.4 Boundary Instability in Moderate Topologies

In datasets with moderate variance overlap (Wine Quality), macroscopic structures are preserved, but micro-level topology diverges. While the core of each K-Means cluster remains stable (high F-Scores), isolated discrepancies emerge exclusively at the spatial margins. Visualized as thin red trajectory flows in the Sankey diagram and justified by hybrid profiles in the PCP, these edge-case misclassifications confirm that the choice between PCA and MDS is critical primarily for boundary points lying between adjacent distributions.



7.5 Diagnostic Identification of Entangled Data

The analytical value of the dashboard is most evident when processing the "User Knowledge" dataset, which lacks inherent, physically separable clusters—as confirmed by the lack of feature isolation in the PCP. In this context, PCA and MDS provide an honest representation of the data's underlying entanglement: they do not manufacture artificial groupings, instead collapsing the data into a single, undifferentiated mass. However, the dashboard's critical diagnostic power lies in the visual analysis of the F-score metric. The high density of red-coded points (indicating a low F-score) serves as a definitive alert for the analyst. This color-coded evidence proves that the projections suffer from extremely low reliability, as they fail to preserve the original high-dimensional neighborhoods within the 2D space. The visual chaos thus transitions from a representation of data to a warning of *topological collapse*.



8 Related Works

The evaluation of Dimensionality Reduction (DR) techniques is a well-documented and evolving challenge within the fields of machine learning and visual analytics. Early approaches relied predominantly on subjective visual inspection to assess projection quality. However, as the dimensionality and complexity of datasets have grown, the literature has strongly shifted toward rigorous, quantitative validation frameworks.

8.1 Foundations of Dimensionality Reduction Quality Metrics

The foundational need to quantify structural distortions was prominently addressed by Venna and Kaski [3], who observed that nonlinear projection methods must guarantee the trustworthiness of proximity relationships to be reliable for exploratory analysis. They introduced early local measures of trustworthiness to penalize visualizations where distant multidimensional points are projected adjacently. This paradigm was mathematically formalized and extended by Lee and Verleysen [4], who introduced a unifying rank-based criteria framework using the co-ranking matrix. Their work systematically defined *Trustworthiness* (penalizing neighborhood intrusions, or false positives) and *Continuity* (penalizing neighborhood extrusions, or false negatives).

Our Positioning: Our system directly implements the mathematical definitions of Trustworthiness and Continuity established by these foundational works to provide a global baseline assessment. However, while Venna & Kaski and Lee & Verleysen primarily focus on the theoretical computation of aggregate scores to rank algorithms, our visual analytics tool democratizes these metrics. We compute them dynamically and display them interactively, serving as a gateway rather than a final conclusion.

8.2 Visualizing Projection Errors

Recognizing that global metrics often obscure local projection failures, Martins et al. [5] proposed techniques to explicitly explain and visualize neighborhood preservation errors. Their work demonstrated that embedding error metrics directly into the visualization—such as coloring points based on local distortion—greatly enhances the interpretability of DR algorithms. Similarly, the comprehensive survey by Espadoto et al. [6] emphasized the critical need for practitioners to understand the dependencies and operational limits of DR techniques through systematic, quantitative mapping.

Our Positioning: We align closely with the visual explanation philosophy of Martins et al. [5] and the quantitative rigor of Espadoto et al. [6]. Our dashboard builds upon this by utilizing continuous color mapping based on relationship metrics directly on the 2D scatterplots. Where our work diverges and innovates is in the semantic application of these spatial errors. Rather than just showing that a topological error occurred, we investigate *how* that error affects downstream data interpretation.

8.3 Precision, Recall, and the Cluster Flow Paradigm

Recently, van der Hoorn et al. [7] introduced a Precision-Recall approach specifically designed to answer the question: *"Why can't I see my clusters?"* They proposed dividing the validation into a relationship phase and a mapping phase, arguing that DR validation must explicitly evaluate how modeled relationships align with expected cluster structures using supervised metrics (Precision and Recall) to track false positives and negatives.

Our Positioning: Our methodology heavily integrates the Precision-Recall logic proposed by van der Hoorn et al. [7] for the relationship phase. However, our proposed visual analytics dashboard advances the state-of-the-art by moving beyond purely spatial and metric-based analysis into **algorithmic clustering disruption**.

While existing literature [5, 7] predominantly evaluates the

projection against abstract ground-truth labels, our tool juxtaposes these abstract labels against data-dependent clusters (computed via N-dimensional and 2D-specific K-Means). The introduction of the *User-2D Perspective* and the *Cluster Agreement Flow* (Sankey diagram) constitutes a novel contribution. It explicitly visualizes the structural flow of data between the PCA 2D space and the Euclidean MDS 2D space, demonstrating not just that distances are distorted, but precisely how these distortions force K-Means to fabricate artificial clusters or fragment natural ones. This provides domain experts with a tangible, interaction-driven mechanism to validate whether the missing visual clusters are an artifact of the DR technique (e.g., PCA's variance maximization) or a true reflection of the multidimensional topology.

9 Conclusion

This study presented a Visual Analytics framework designed to demystify the structural distortions inherent in Dimensionality Reduction (DR). By shifting the focus from purely aesthetic spatial embeddings to a rigorous, metric-driven diagnostic approach, we demonstrated that the choice between PCA and MDS is a critical decision that dictates the reliability of downstream data interpretation. Our findings confirm that linear techniques like PCA are highly susceptible to "topological collapse" in entangled datasets (e.g., *User Knowledge*), often manufacturing artificial clusters or failing to preserve local neighborhoods. Conversely, the dashboard highlights how Euclidean MDS, through iterative stress minimization, offers a more faithful representation of non-linear manifolds, as evidenced by superior Trustworthiness and Continuity scores. The integration of the **Cluster Agreement Flow (Sankey)** and **F-Score color mapping** allows analysts to distinguish between "natural" data groupings and "algorithmic artifacts." When projections fail, the dashboard provides an explicit warning of low reliability, effectively preventing domain experts from drawing false conclusions from distorted 2D projections and establishing a new standard for the validation of visual embeddings.

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